Transformări ortogonale

(1) Reflectori Householder

$$H = I_n - \beta u \cdot u^T \qquad u \in \mathbb{R}^n,\ u \neq 0$$

$$\beta = \frac{2}{u^T \cdot u}$$

Prop 1 H simetrică $(H = H^T)$

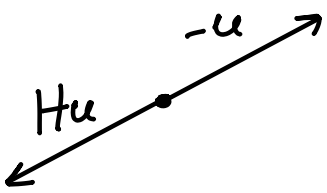
$$H^T = (I_n - \beta u \cdot u^T)^T = I_n - \beta \cdot (u \cdot u^T)^T =$$
$$= I_n - \beta \cdot u \cdot u^T = H$$

Prop 2 $H = H^{-1}$

$$H^2 = H \cdot H = (I_n - \beta u \cdot u^T)(I_n - \beta u \cdot u^T) =$$
$$= I_n - 2 \cdot \beta u \cdot u^T + \beta^2 \cdot u \cdot \underbrace{u^T \cdot u}_{\frac{2}{\beta}} \cdot u^T =$$
$$= I_n - 2 \cdot \beta \cdot u \cdot u^T + \beta \cdot u \cdot \frac{2}{\beta} \cdot u^T$$
$$= I_n - 2\beta u \cdot u^T + 2\beta \cdot u \cdot u^T = I_n$$

$H = H^{-1} = H^T \Rightarrow H$ ortogonală

Prop 3 $H \cdot u = -u$



$$H \cdot u = (I_n - \beta u \cdot u^T) \cdot u = u - \beta \cdot u \cdot u^T \cdot u$$
$$= u - \beta \cdot u \cdot \frac{2}{\beta} = u - 2u = -u$$

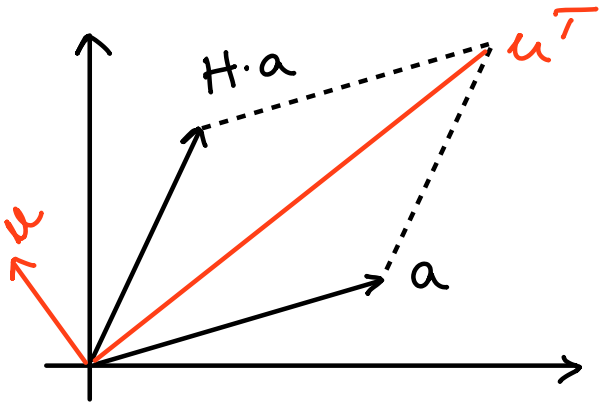
Prop 4 $a - Ha = \rho u$

$$H \cdot a = (\mathcal{I}_n - \beta u \cdot u^T) \cdot a = a - \beta u \cdot u^T a$$

$$\rho = \beta \cdot u^T a \ (\text{scalar})$$

$$\Rightarrow H \cdot a = a - \rho \cdot u \Rightarrow$$

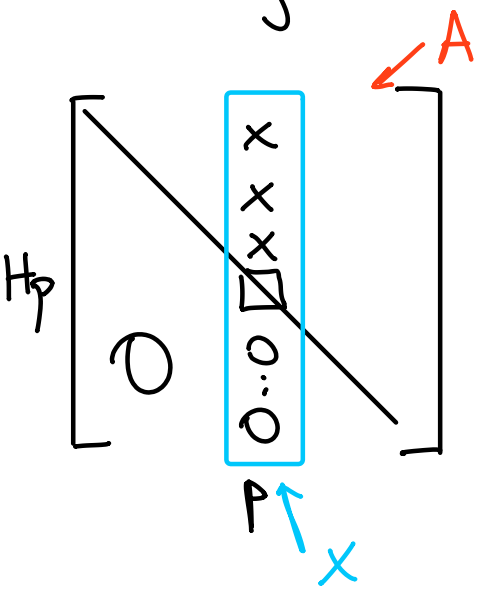
$$\Rightarrow a - Ha = \rho u$$



$$A = \begin{bmatrix} x & x & \cdots \\ 0 & x & \ddots \\ 0 & 0 & \ddots \\ \vdots & \vdots & \ddots \\ 0 & 0 & \cdots \end{bmatrix}$$

$a = A(:,1)$ = coloana 1 din matricea A

$H \cdot a = \rho e_1$



$$H_p = \mathcal{I}_n - \beta u_p \cdot u_p^T$$

$$u_p = \begin{cases} 0 & i = 1:p-1 \\ x_p + \nabla & i = p \\ x_i & i = p+1:n \end{cases}$$

$x = A(:,p)$

= coloana p din mat. A

$$H_p \cdot x = (\mathcal{I}_n - \beta u_p \cdot u_p^T) \cdot x =$$

$$= x - \beta (u_p^T \cdot x) \cdot u = x - u = \begin{cases} x_i & i = 1:p-1 \\ -\nabla_p & i = p \\ 0 & i = p+1:n \end{cases}$$

$$H \cdot a = \rho e_1$$

$$\|Ha\| = \|a\| = \rho \cdot \|e_1\| = \rho$$

$$\sigma_p = \|x(p:n)\| = \sqrt{\sum_{i=p}^{n} x_i^2}$$

$\uparrow$ $\operatorname{sgn}(x_p)$

$$u_p^T \cdot x = \sum_{i=1}^{n} u_{pi} \cdot x_i = x_p(x_p + \sigma_p) + \underbrace{\sum_{i=p+1}^{n} x_i^2}_{\sigma_p^2 - x_p^2} =$$

$$= x_p^2 + \sigma_p x_p + \sigma_p^2 - x_p^2 = \sigma_p(\sigma_p + x_p)$$

$$u_p^T \cdot u_p = (x_p + \sigma_p)^2 + \sigma_p^2 - x_p^2 = 2\sigma_p^2 + 2\sigma_p x_p = 2\sigma_p(\sigma_p + x_p)$$

$$\rightarrow \beta = \frac{1}{\sigma_p(\sigma_p + x_p)} \qquad \beta(u_p^T \cdot x) = 1$$

$$H_p \cdot x = x - u = \begin{cases} x_i & i = 1:p-1 \\ -\sigma_p & i = p \\ 0 & i = p+1:n \end{cases}$$

$$H_p \cdot y = y \qquad y = A(:, j) \qquad j = 1:p-1$$

$$y = \begin{cases} y_i & i = 1:j \\ 0 & i = j+1:n \end{cases}$$

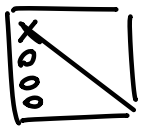
$$H_p \cdot y = (\mathcal{I}_n - \beta u \cdot u^T) y = y - \gamma \cdot u$$

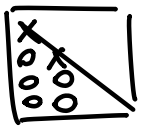
$y = A(:, j) \quad j = p+1 : m$

$\beta_y = Bu^T y$ (scalar)

$H_p \cdot y = y - \beta_y \cdot u$

$A^{(0)} = A$

$A^{(1)} = H_1 A^{(0)} = H_1 A$ 

$A^{(2)} = H_2 A^{(1)} = H_2 \cdot H_1 A$ 

$\vdots$

$A^{(n-1)} = H_{n-1} A^{(n-2)} = H_{n-1} H_{n-2} H_{n-3} \dots H_2 H_1 A$

$R = A^{(n-1)} = H_{n-1} H_{n-2} H_{n-3} \dots H_2 H_1 A$

$R = Q^T \cdot A \Rightarrow Q = (H_{n-1} H_{n-2} H_{n-3} \dots H_2 H_1)^T$

$Q = H_1 H_2 \dots H_{n-1}$

A, B ortogonale

$(A \cdot B)^T AB = B^T A^T \cdot AB = B^T \cdot B = I_n$

$A^T A = I_n$

reflectă față de planul $\perp$ pe u

$H = I_n - \beta u \cdot u^T$

(2) Rotații Givens

$$G_{kl} = \begin{bmatrix} I_{k-1} & & & & \\ & c & & s & \\ & & I_{n-l-k-2} & & \\ & -s & & c & \\ & & & & I_{n-l-1} \end{bmatrix} \begin{matrix} \\ k \\ \\ l \\ \\ \end{matrix}$$

$$c^2 + s^2 = 1$$

Exercițiu $\rightarrow G_{kl} \cdot G_{kl}^T = I_n \Rightarrow G_{kl}$ ortogonal

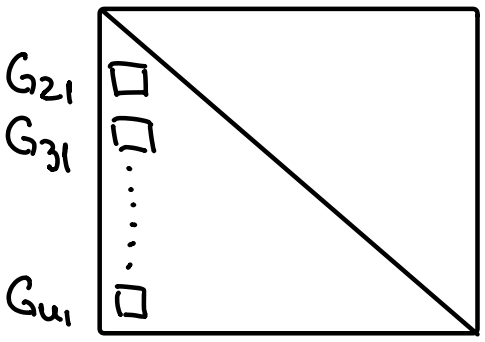
$$G_{kl} \cdot A = \begin{matrix} k \\ l \end{matrix} \begin{bmatrix} c & & s \\ & I & \\ -s & & c \end{bmatrix} \begin{matrix} k \\ \\ l \end{matrix} \begin{bmatrix} a_{kk} & \cdots & a_{kc} \\ \vdots & & \vdots \\ a_{lk} & \cdots & a_{ll} \end{bmatrix}$$

$a_{lk} \rightarrow 0$

$$\begin{cases} -s a_{kk} - c a_{lk} = 0 \leftarrow \text{se cere} \\ s^2 + c^2 = 1 \end{cases}$$

$$s = \frac{a_{lk}}{a_{kk}} \cdot c \quad \Rightarrow \quad c = \frac{\pm a_{kk}}{\sqrt{a_{kk}^2 + a_{lk}^2}}$$

$$s = \frac{\pm a_{lk}}{\sqrt{a_{kk}^2 + a_{lk}^2}}$$



$\tilde{G}_1 = G_{21} G_{31} \cdots G_{n1} \rightarrow$ coloana 1

$\tilde{G}_2 = G_{32} G_{42} \cdots G_{n2} \rightarrow$ coloana 2

$\vdots$

$\tilde{G}_{n-1} = G_{nn-1}$

$Q = \tilde{G}_1 \tilde{G}_2 \tilde{G}_3 \cdots \tilde{G}_{n-1} \rightarrow$ ortogonală

Implementare

$$A=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad G=\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

$$G\cdot A=\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$

$$G=\begin{pmatrix} \frac{a}{\sqrt{a^2+c^2}} & \frac{c}{\sqrt{a^2+c^2}} \\ \frac{-c}{\sqrt{a^2+c^2}} & \frac{a}{\sqrt{a^2+c^2}} \end{pmatrix}$$

$$G_{13}=\begin{pmatrix} \cos & 0 & -\sin \\ 0 & 1 & 0 \\ \sin & 0 & \cos \end{pmatrix}$$

$$G_{uu}^t\cdot G_{uu-1}^t\cdot\ldots\cdot G_{1u}^t\cdot G_{1u-1}^t\cdot\ldots\cdot G_{12}^t\cdot A=R$$

$$Q=G_{12}\ldots G_{uu}^t$$

ex

$$A=\begin{pmatrix} \overset{a}{3} & \overset{b}{1} & 3 \\ c\,4 & 2\,d & 0 \\ 1 & 1 & 4 \end{pmatrix} \qquad G^t=\begin{pmatrix} 3/5 & 4/5 & 0 \\ -4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

! A diagonal dominantă (elementele de pe diagonală sunt cele mai mari)

(3) Algoritmul Gram Schmidt

$$[a_1\ a_2 \dots a_n] = [q_1\ q_2 \dots q_n] \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ 0 & r_{22} & \dots & r_{2n} \\ 0 & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ 0 & \vdots & \dots & r_{nn} \end{bmatrix}$$

Coloana 1

$r_{11} q_1 = a_1 \Rightarrow r_{11} = \|a_1\|_2$ $\quad (\|q_1\|_2 = 1)$

$$q_1 = \frac{a_1}{r_{11}}$$

Coloana 2

$$r_{12} q_1 + r_{22} q_2 = a_2$$

$q_1^T \mid r_{12} q_1 + r_{22} q_2 = a_2 \Rightarrow r_{12} \underbrace{q_1^T q_1}_{1} + r_{22} \underbrace{q_1^T q_2}_{0} = q_1^T a_2$

$$r_{12} = q_1^T a_2$$

$$r_{22} q_2 = a_2 - r_{12} q_1$$

$$r_{22} = \|a_2 - r_{12} q_1\|_2$$

$$q_2 = \frac{a_2 - r_{21} q_1}{r_{22}}$$

Coloana k

$$\sum_{i=1}^{k} r_{ik} q_i = a_k$$

$q_i^t \mid \sum_{i=1}^{k} r_{ik} q_i = a_k$

$j = 1 : k-1$ $\qquad r_{ik} = q_j^t a_k$

$$r_{kk} = \left\| a_k - \sum_{i=1}^{k-1} r_{ik} q_i \right\|_2$$

$$q_k = \frac{1}{r_{kk}} \left(a_k - \sum_{i=1}^{k-1} r_{ik} q_i \right)$$

Implementare

$$A = (a_1\ a_2 \dots a_n) \longrightarrow Q = (q_1\ q_2 \cdots q_n)$$

$$q_1 = a_1$$

$$q_2 = a_2 - \text{pr}_{q_1} a_2 = a_2 - \frac{\langle a_2, q_1 \rangle}{\langle q_1, q_1 \rangle} \cdot a_1$$

$$q_3 = a_3 - \text{pr}_{q_1} a_3 - \text{pr}_{q_2} a_3$$